

Constituency Tests for Concerned Representation in Minimal Agents

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Abstract

Arc 1 of the maintained-concern program showed that a minimal homeostatic agent can detect boundary staleness, allocate costly null probes, saturate after identification, and re-engage after regime shifts. It also found a precise ceiling: under a shared-head, null-only intervention regime, better probe policy no longer closes role-specific mediated identifiability. This paper opens Arc 2A by asking whether the next intervention problem is syntactic: can a concerned agent discover which parts of a world belong together causally?

We introduce the **Concerned Shape Grammar**, a symbolic benchmark inspired by constituency tests from cognitive work on geometric shape syntax. A visible six-part shape can have the same surface but different hidden parse trees. Causal roles such as shield, poison, repair, core, food, and trap interact only when they are bound inside the same constituent. The agent must decide whether to pay for an intervention that reveals the parse, and it must avoid probing low-concern ambiguity that does not affect viability.

In a 200-trial deterministic design pilot, a 5,000-trial symbolic Modal sweep, and a learned 5-seed Modal sweep, the concern-gated syntax agent is the only agent family that passes the full gate. The learned agent receives candidate parses as visible hypotheses but never receives the hidden true parse at test time. It learns a tree-binding parse interpreter, an intervention policy, and a capped low-concern calibration guard. Across five Modal seeds, it reaches high-concern parse accuracy 1.000, action accuracy 1.000, low-concern probe rate 0.202, and gate pass rate 1.000. Shortcut reward, no-tree planning, and restless tree inquiry all fail for different anti-cheat reasons. The result is still not a claim about pixel-level perception or human cognition. It is an accepted Phase 2A learned-mechanism surface: **reward is not syntax, compression is not syntax, and uncertainty reduction is not concerned inquiry.**

1. Why Arc 2A Exists

The Metric Stack of Concern ended with a positive mechanism and a ceiling. The positive mechanism was a detect-allocate-saturate-re-engage cycle for self/world attribution in minimal homeostatic agents. The ceiling was equally important: under null-only intervention and shared mediated heads, the agent could predict total world response while failing to identify role-specific mediated components.

That ceiling has two sides. Arc 2B studies the body side: what architectures can express the required distinctions? Arc 2A studies the intervention side: what actions make the world's hidden grammar visible?

The motivation comes from a nearby but distinct empirical result. Revencu, Pajot, and Dehaene (2026) argue that adult geometric-shape representations show syntactic structure. Their key methodological move is not merely to show that shape complexity predicts behavior. They replace compression proxies with constituency tests: structural ambiguity, subtree facilitation, and syntactic movement. They also report that current neural networks can partially solve match/deviant tasks without showing the same syntactic effects.

This paper imports that methodological lesson into the concern program. The claim is not that minimal agents have human visual syntax. The claim is that the Metric Stack needs an analogous gate:

█ Performance is not constituency. Compression is not constituency. Uncertainty reduction is not

concerned inquiry.

2. Benchmark

Each trial has a six-part visible shape. The visible role sequence is fixed, but two hidden parse trees can organize the same surface differently. A parse is a pair of high-level constituents, each containing three leaves.

Examples of parse candidates:

```
repeat_concat      = (0,1,2) | (3,4,5)
hooked_repeat      = (0,1,3) | (2,4,5)
alternating_bind   = (0,2,4) | (1,3,5)
edge_core          = (0,4,5) | (1,2,3)
```

Causal roles interact only when they belong to the same constituent:

Role pair	Causal rule
shield + poison	shield reduces poison damage only within a subtree
repair + core	repair reduces core damage only within a subtree
food + trap	trap contaminates food only within a subtree
signal + ornament	structural ambiguity with no viability consequence

The low-concern case is essential. Without it, a generic uncertainty reducer could look successful simply by probing every ambiguity.

3. Interventions

The intervention language contains five operations:

Intervention	Purpose
null	no information, no cost
pair_probe	test whether the causal role pair is in the same subtree
distractor_pair_probe	matched non-informative pair probe
high_constituent_move	test which high-level group moves as a unit
role_ablation	observe the viability signature of the causal constituent

The central selector, `concerned_syntax`, chooses an intervention by expected parse information weighted by the viability gap, minus intervention cost. It has a hard no-restless-inquiry threshold: if the parse gap is below 0.10, it chooses `null`.

4. Selectors

Selector	What it tests
null_policy	behavior without inquiry
flat_valence	roles without constituency
compression_proxy	shortest parse without intervention
uncertainty_only	information gain without concern
concerned_syntax	information gain gated by viability relevance

This is the anti-cheat surface. A selector can be good at action while failing parse. It can recover parse while probing too much. It can choose a compressed parse while missing the causally relevant one.

5. Pilot Result

Local command:

```
python3 -m experiments.concerned_syntax.benchmark \
  --trials 200 --seed 20260616 \
  --out artifacts/concerned_syntax/pilot.json \
  --report experiments/concerned_syntax/results/pilot_2026_06_16.md
```

Summary:

Selector	Parse high	Action	Subtree	High probe	Low probe	Mean regret	Gate
compression_proxy	0.523	0.890	0.550	0.000	0.000	0.055	fail
concerned_syntax	1.000	1.000	0.805	1.000	0.000	0.001	PASS
flat_valence	0.000	0.920	0.540	0.000	0.000	0.074	fail
null_policy	0.523	0.890	0.550	0.000	0.000	0.055	fail
uncertainty_only	1.000	1.000	1.000	1.000	1.000	0.000	fail

The result has the desired diagnostic pattern:

1. flat_valence has high action accuracy but fails parse and subtree gates.
2. compression_proxy sometimes guesses the right parse, but does not intervene and fails the mechanistic gate.
3. uncertainty_only recovers parse perfectly but probes all low-concern ambiguity, failing no-restless-inquiry.
4. concerned_syntax probes every high-concern ambiguity, avoids low-concern probing, and preserves action.

6. Discovery-Regime Status

This is a regime transition relative to Arc 1. Arc 1 represented concern as viability prediction, valence, self/

world attribution, probe value, and component identifiability. Arc 2A adds a new artifact type: **causal constituency under concern**.

That makes the result discovery-leaning, but still benchmark-level. The pilot does not show that a neural agent has learned syntax. It shows that the repo now has a task surface on which syntax, compression, uncertainty, and concern can dissociate.

7. Modal Multi-Seed Sweep

The multi-seed sweep was run remotely through Modal:

```
doppler --scope /Users/jawaun/superoptimizers run -- \
  uvx --python 3.12 --from modal modal run \
  experiments/concerned_syntax/modal_concerned_syntax_sweep.py \
  --trials 1000
```

The sweep used five seeds and 1,000 shape trials per seed. Raw JSON remains local under `artifacts/concerned_syntax/`; the public report is `experiments/concerned_syntax/results/modal_sweep_2026_06_16.md`.

Summary:

Selector	Parse high	Action	Subtree	High probe	Low probe	Mean regret	Gate pass rate
compression_proxy	0.560	0.891	0.583	0.000	0.000	0.048	0.000
concerned_syntax	1.000	1.000	0.808	1.000	0.000	0.003	1.000
flat_valence	0.000	0.876	0.503	0.000	0.000	0.066	0.000
null_policy	0.560	0.891	0.583	0.000	0.000	0.048	0.000
uncertainty_only	1.000	1.000	1.000	1.000	1.000	0.000	0.000

This replicates the design-pilot pattern across seeds. The key result is not that `concerned_syntax` has the best reward; `uncertainty_only` has zero regret too. The key result is that only concern-weighted syntax passes both the positive inquiry gate and the no-restless-inquiry gate.

8. Learned-Agent Gate

The next gate removes the hand-coded selector. Candidate parses remain visible as hypotheses, as in a structural-ambiguity task, but the true parse is hidden at test time. A small learned agent trains three binary components:

- a concern-gated intervention policy;
- a parse interpreter that binds pair-probe observations to candidate constituents;
- a shortcut action head used as a reward-only control.

The tree-binding body receives candidate-subtree equality features for the probed pair. The no-tree control receives the same surface roles, pair identity, concern weight, and probe observation, but not the candidate-binding features needed to attach that observation to a constituent. The guarded learner also gets a

deterministic 20% low-concern calibration budget. This is below the 0.25 anti-restless cap and prevents syntax maintenance from failing exactly at the subtree threshold.

Remote command:

```
doppler --scope /Users/jawaun/superoptimizers run -- \
  uvx --python 3.12 --from modal modal run \
  experiments/concerned_syntax/modal_learned_agents_sweep.py \
  --train-trials 3000 --test-trials 1200 --epochs 90
```

Summary:

Agent	Parse high	Action	Subtree	High probe	Low probe	Gate
guarded syntax	1.000	1.000	0.797	1.000	0.202	PASS
no-tree planner	0.492	0.875	0.494	1.000	0.000	fail
restless tree	1.000	1.000	1.000	1.000	1.000	fail
shortcut reward	0.494	0.880	0.495	0.000	0.000	fail

The full public report also records mean probe cost and regret.

This is the first learned Phase 2A result. It shows that a learner can pass the concerned-syntax gate without hidden parse access, but only when it has both the tree-binding feature needed for causal constituency and the formal guard needed to prevent restless low-concern inquiry.

The next version should add richer learned agents:

- tree-structured model
- flat MLP baseline
- object-slot baseline
- learned intervention policy
- role/parse held-out generalization
- neural anti-cheat probes for parse, subtree, and intervention usefulness

9. Limitations

The current benchmark is symbolic. The learned agent receives vectorized surface roles and candidate parse hypotheses, not pixels. It does not test continuous control or human subjects. The intervention language is provided, not invented from raw motor primitives. The parser candidates are small and known to the evaluator. The point of this first paper is to define and pass a minimal learned acceptance surface before larger compute.

The most important limitation is also the next step: the agent should learn the intervention language and parse representation from richer generated shapes, not merely bind provided candidate parses to provided probe observations.

10. Conclusion

Arc 2A inserts a new layer into the maintained-concern ladder:

difference -> geometry -> syntax -> salience -> valence
-> action -> attribution -> maintenance

The results support a narrow methodological claim: concerned syntax needs its own tests. Reward, compression, uncertainty, action accuracy, and even syntax without concern can all dissociate from causal constituency under maintained concern. The learned-agent sweep shows that the gate is passable without hidden parse access, but only when tree binding, intervention, and formal concern gating are present together.

References

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